

Problem

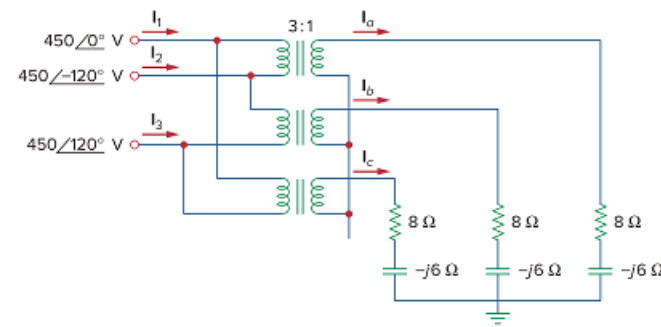
Figure 13.135 on the next page shows a three-phase transformer that supplies a Y-connected load.

(a) Identify the transformer connection.

(b) Calculate currents I_2 and I_c .

(c) Find the average power absorbed by the load.

Figure 13.135



Step-by-step solution

Step 1 of 9

Refer to figure 13.135 in the textbook.

Redraw the circuit as shown in figure 1.

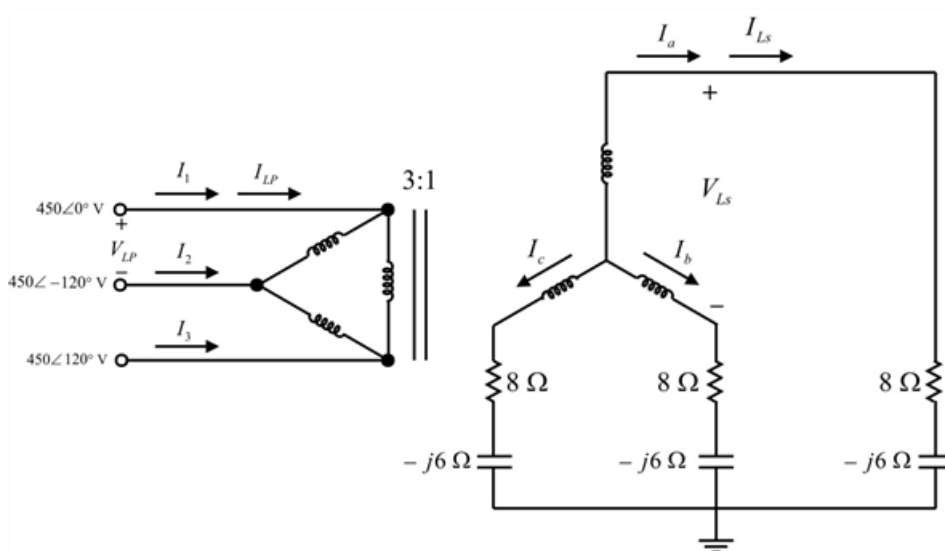


Figure 1

Step 2 of 9

(a)

Consider the circuit shown in figure 1.

The primary side is Δ -connected, whereas the secondary side is Y-connected.

Therefore, the circuit is **three-phase Δ -Y transformer**.

Step 3 of 9

(b)

Consider the following relation:

$$1:n=3:1$$

$$\frac{1}{n}=3$$

$$n=\frac{1}{3}$$

Step 4 of 9

Calculate the secondary line voltage V_{Ls} .

$$V_{Ls} = n\sqrt{3}V_{LP}$$

Consider the circuit shown in figure 1.

Substitute $450\angle 0^\circ$ for V_{LP} , and $\frac{1}{3}$ for n .

$$\begin{aligned} V_{Ls} &= \frac{1}{3}\sqrt{3}(450\angle 0^\circ) \\ &= 86.6\angle 0^\circ \text{ V} \end{aligned}$$

Step 5 of 9

Calculate the secondary line current I_{Ls} .

$$\begin{aligned} I_{Ls} &= \frac{V_{Ls}}{8-j6} \\ &= \frac{V_{Ls}}{10\angle -36.87^\circ} \end{aligned}$$

Substitute $86.6\angle 0^\circ$ for V_{Ls} .

$$\begin{aligned} I_{Ls} &= \frac{86.6\angle 0^\circ}{10\angle -36.87^\circ} \\ &= \frac{86.6}{10}\angle (36.87^\circ) \\ &= 8.66\angle 36.87^\circ \text{ A} \end{aligned}$$

Step 6 of 9

Calculate the primary line current I_{lp} .

$$I_{lp} = n\sqrt{3}I_{Ls}$$

Substitute $8.66\angle 36.87^\circ$ for I_{Ls} , and $\frac{1}{3}$ for n .

$$\begin{aligned} I_{lp} &= \frac{1}{3}\sqrt{3} \times 8.66\angle 36.87^\circ \\ &= 4.9998\angle 36.87^\circ \text{ A} \end{aligned}$$

Step 7 of 9

Consider the circuit shown in figure 1.

The line current I_1 is,

$$I_1 = I_{lp} = 4.9998\angle 36.87^\circ \text{ A}$$

The line current I_2 is,

$$\begin{aligned} I_2 &= I_1\angle -120^\circ \\ &= 4.9998\angle -83.13^\circ \text{ A} \end{aligned}$$

Therefore, the value of current I_2 is $\boxed{4.9998\angle -83.13^\circ \text{ A}}$.

Step 8 of 9

Consider the circuit shown in figure 1.

The line current I_a is,

$$I_a = I_{Ls} = 8.66\angle 36.87^\circ \text{ A}$$

The line current I_c is,

$$\begin{aligned} I_c &= I_a\angle 120^\circ \\ &= 8.66\angle 156.87^\circ \text{ A} \end{aligned}$$

Therefore, the value of current I_c is $\boxed{8.66\angle 156.87^\circ \text{ A}}$.

Step 9 of 9

(c)

The average power absorbed by the load is,

$$P = 3(I_a)^2 R$$

Substitute 8.66 for I_a , and 8 for R .

$$\begin{aligned} P &= 3 \times (8.66)^2 \times 8 \\ &= 1.8 \text{ kW} \end{aligned}$$

Therefore, the average power absorbed by the load is $\boxed{1.8 \text{ kW}}$.